

LOAN TRACKING SYSTEM

USER MANUAL

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Introduction

The Loan Tracking System is a personal finance web application designed for a single user (e.g., David Pasumbal, who is represented in the app interface) to track his personal loans, debts, installment payments, and shared group expenses.

System Overview

The software allows the user to easily manage:

- **Who owes them money:** what they have lent out to individual people or groups.
- **What they owe others:** what they have borrowed from individual contacts.
- **Transaction Types:** Supports Straight Expenses (one-time loans), Installment Expenses (loans with structured recurring weekly/monthly schedules), and Group Expenses (expenses shared and split among a group).

System Requirements

To run and access the application, the following specifications are required:

- **Client / Supported Browsers:** Any modern web browser with HTML5, CSS3, and ES6+ JavaScript support (Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge).
- **Necessary Backend Software:**
 - **Java Runtime:** Java 21 or higher.
 - **Build Tool:** Maven (or using the included Maven Wrapper `./mvnw`).
 - **Framework:** Spring Boot 3.5.x.
- **Database:** PostgreSQL server (running locally or remotely, configured via `src/main/resources/application.properties`).

Account Setup

- **No Authentication Layer:** As specified in the system design assumptions, the application operates in a single-user environment and does not contain authentication, sign-up flows, login screens, or password reset mechanisms.
- **Predefined Identity:** The active user is automatically represented in the database as a primary contact PERSON record (with ID 11111111-1111-1111-1111-111111111111).

You do not need to create an account; launching the backend and opening **index.html** grants immediate access to the dashboard under this profile.

User Roles & Permissions

- **Single-Role System:** There are no separate access levels (such as Admin, Teller, or Auditor).
- **Permissions:** The single active user has full Read, Write, Update, and Delete (CRUD) permissions across all resources in the system: Creating and managing individual contacts (PERSON) and circles (GROUP_ENTITY). Creating, editing, and deleting loan entries (LOAN_ENTRY). Creating and updating payment transactions (PAYMENT). Tracking installment schedules (INSTALLMENT_PLAN / INSTALLMENT_TERM) and splitting group expenses (PAYMENT_ALLOCATION).

Getting Started

Part 1: PostgreSQL & pgAdmin 4 Setup

Before running the application, you need to configure your database server and set up the specific user and database schemas.

Step 1: Create the Database User Role

1. Open **pgAdmin 4** and connect to your local server.
2. In the left-hand Browser tree, expand your server, right-click on **Login/Group Roles**, hover over **Create**, and select **Login/Group Role....**
3. Configure the following settings:
 - **General Tab:** In the **Name field**, type exactly: `loan_user`
 - **Definition Tab:** In the **Password field**, type: `password`.
 - **Privileges Tab:** Toggle **Can login?** To **Yes**. You can also toggle **Superuser** to **Yes** for local academic/development environments to avoid permission issues.
 - Click **Save**.

Step 2: Create the Database

1. Right-click on **Databases** in the left-hand Browser tree, hover over **Create**, and select **Database....**
2. Configure the following:
 - **Database:** `loan_tracker`
 - **Owner:** Select `loan_user` from the dropdown list.
3. Click **Save**.

(Note: Spring Boot is configured with Hibernate/JPA to automatically generate tables using the `schema.sql` script once it connects, so you do not need to manually write tables in pgAdmin.)

Part 2: Starting the Spring Boot Backend

With the database up and listening, you can boot up your Java server via the terminal.

Step 1: Open the Terminal in the Project Directory

1. Open your project in VS Code.
2. Ensure your terminal is pointing exactly to your inner backend directory with the use of `cd (...\\backend\\backend)`

```
PS C:\Users\sabri\Downloads\backend> cd C:\Users\sabri\Downloads\backend\backend
PS C:\Users\sabri\Downloads\backend\backend> █
```

Step 2: Run the Maven Wrapper Command

Execute the Spring Boot plugin run command depending on your operating system:

- On Windows (PowerShell / CMD):

```
None
./mvnw spring-boot:run
```

- On macOS / Linux:

```
None
./mvnw spring-boot:run
```

Step 3: Verify the Backend is Active

Watch the console output. You should see the Spring banner display, Hibernate logs initializing the schema, and a final line resembling: `[INFO] Started BackendApplication in X.XXX seconds (process running for X.XX)`

```
PS C:\Users\sabri\Downloads\backend\backend> ./mvnw spring-boot:run
2026-05-23T10:55:49.012+08:00 INFO 25968 --- [backend] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on
port 8080 (http) with context path '/'
2026-05-23T10:55:49.019+08:00 INFO 25968 --- [backend] [           main] c.l.backend.BackendApplication       : Started BackendApp
lication in 4.5 seconds (process running for 4.779)
-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure spring.jpa.open-i
n-view to disable this warning
2026-05-23T10:55:49.012+08:00 INFO 25968 --- [backend] [           main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on
port 8080 (http) with context path '/'
2026-05-23T10:55:49.019+08:00 INFO 25968 --- [backend] [           main] c.l.backend.BackendApplication       : Started BackendApp
lication in 4.5 seconds (process running for 4.779)
```

Important: Keep this terminal window open. Closing it will terminate the backend server.

(Ctrl + c to terminate)

Part 3: Accessing the Frontend Interface

Since the frontend consists of standard, flat client files, launching it is instant.

Open the Main Entry Point

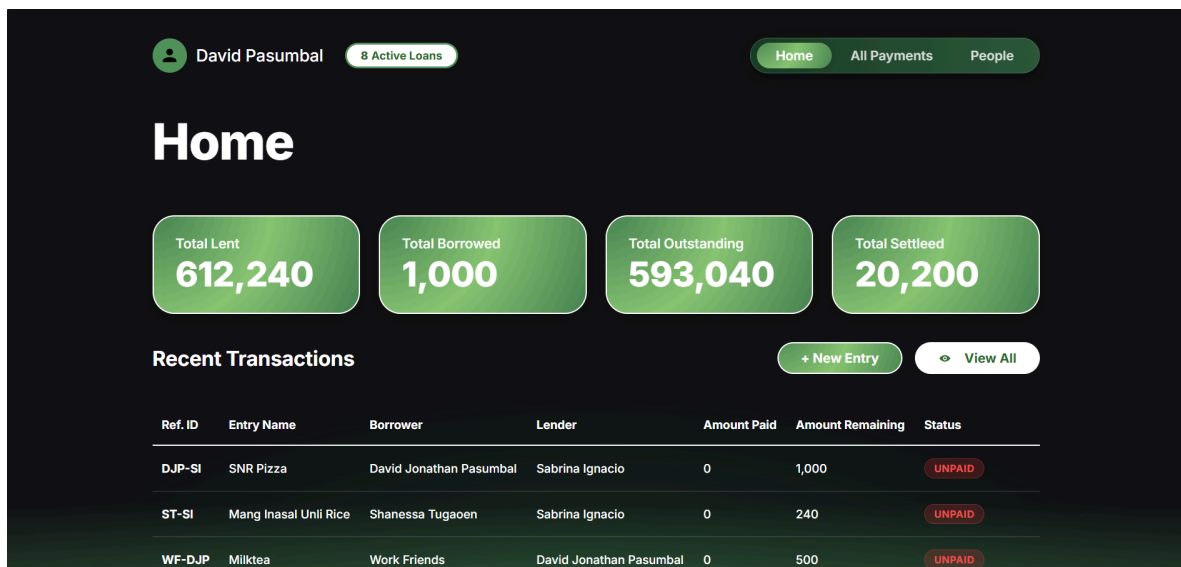
1. Open your file explorer or your VS Code sidebar.
2. Navigate to the project's root folder (one level above the backend folder).
3. Find the file named **index.html**.
4. **Double-click** it to launch it directly in your default web browser (Chrome, Edge, Firefox, etc.).

Dashboard Navigation

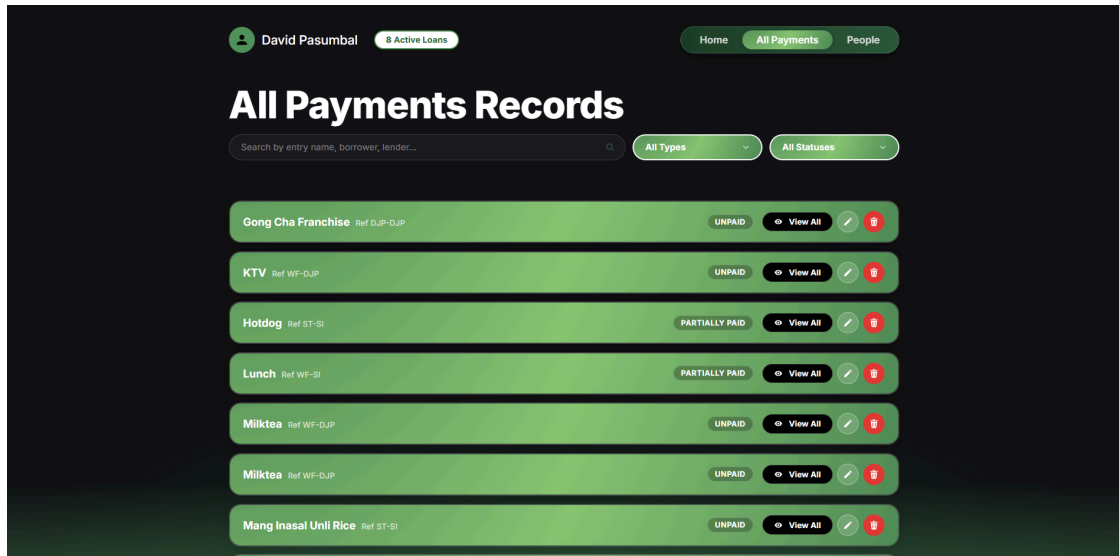
The workspace uses a navigation header with user profile details and a navigation menu to browse different sections of the app. The bottom of the homepage features quick actions to add new loan records or view all active ledger entries.

Main interface

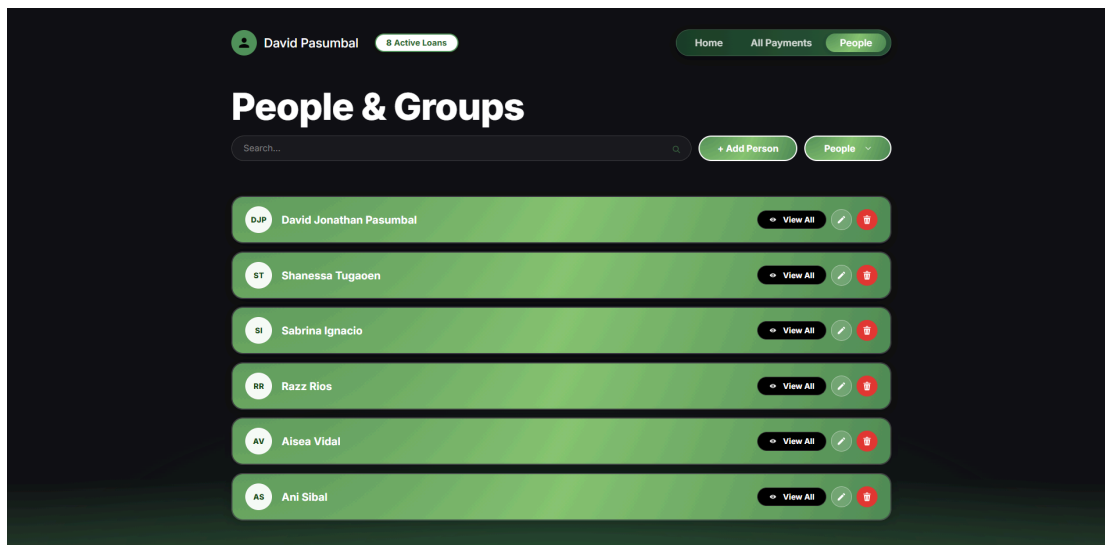
- **Home:** It provides an immediate high-level summary of your net financial status through four distinct metrics card widgets, lists your 5 most recent transactions, and contains action buttons to create new entries.



- **All payments:** This page displays a comprehensive list of all loan entries. It features advanced search and filter controls where you can query entries by name, contact, or reference ID, and filter the list by **Transaction Type** (Straight, Installment, Group) or **Payment Status** (Unpaid, Partially Paid, Paid). Clicking any record redirects you to its payment history.



- **People:** This is the contact directory. This interface lets you manage contacts and groups. It has a toggle dropdown to switch between:
 - **People view:** Add, edit, or delete contacts. Expanding a contact's card displays their specific lending/borrowing history.
 - **Groups view:** Manage expense groups (e.g. shared bills). You can add or remove group members and view entries split among the group.



Key Metrics Widget (Dashboard)

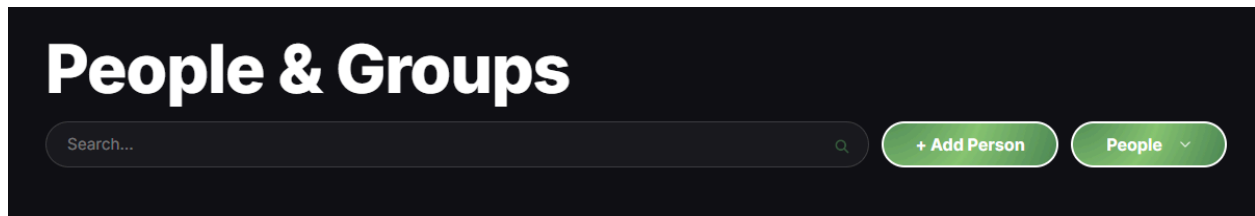
Located at the top of the Home dashboard, these cards give you a snapshot of your cash flow:

- **Total Lent:** The gross sum of all money you have loaned out to others (representing assets you expect to collect).
- **Total Borrowed:** The gross sum of all money you have borrowed from others (representing liabilities you need to pay back).
- **Total Outstanding:** The net outstanding balance ($|\text{Lent Outstanding} - \text{Borrowed Outstanding}|$). The widget is color-coded to reflect your net position:
 - Green: You are a net creditor (others owe you more than you owe them).
 - Red: You are a net debtor (you owe others more than you are owed).
- **Total Settled:** The total accumulated amount that has been paid off across all active or completed loan entries.

Borrower/Client Management

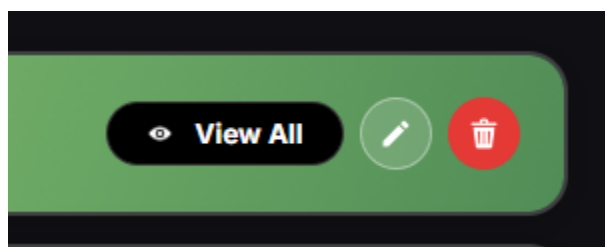
Before any loan can be tracked, you need to manage the clients/people taking out the loans

Adding a New Borrower/Client



1. Navigate to the **People** Tab.
2. Click the **+ Add Person** (make sure the tab selector is on People).
3. Required Fields
 - a. First Name: First name of the client.
 - b. Last Name: Last name of the client
4. Optional Fields
 - a. Contact Information: Phone number, email, or other contact details/
 - b. Notes: Extra information such as address, preferences, etc.
5. Click **Save**. Upon saving, a unique ID is automatically generated by the database for that client.

Editing and Updating Profiles

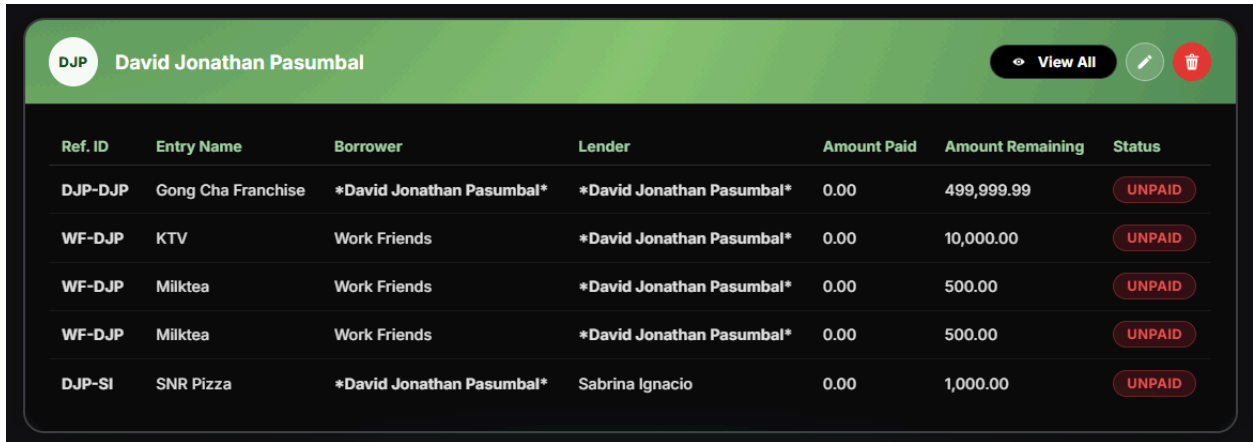


If a client changes their address, phone number, or other details:

1. Locate the person's card under the **People** tab list.
2. Click the Edit Button on the right side of their card.
3. Update their details in the modal.
4. Click **Save** to apply the changes

Viewing Borrower History

The system automatically highlights the client's name inside the Borrower or Lender columns, helping you instantly see whether they acted as the borrower or lender for each entry.



Ref. ID	Entry Name	Borrower	Lender	Amount Paid	Amount Remaining	Status
DJP-DJP	Gong Cha Franchise	*David Jonathan Pasumbal*	*David Jonathan Pasumbal*	0.00	499,999.99	UNPAID
WF-DJP	KTV	Work Friends	*David Jonathan Pasumbal*	0.00	10,000.00	UNPAID
WF-DJP	Milktea	Work Friends	*David Jonathan Pasumbal*	0.00	500.00	UNPAID
WF-DJP	Milktea	Work Friends	*David Jonathan Pasumbal*	0.00	500.00	UNPAID
DJP-SI	SNR Pizza	*David Jonathan Pasumbal*	Sabrina Ignacio	0.00	1,000.00	UNPAID

To inspect a client's past and current loans:

1. Find the client's card under the **People** tab.
2. Click the **View All** button on their card.
3. The card will expand to show a table listing all loan entries associated with that client.

Loan Management

This section covers how to manage loans– from creation, through the amortization schedule, to activation and final closure.

Creating a New Loan

Basic Information About the Loan

The screenshot shows a 'New Loan Entry' form. At the top, there's a user profile for 'David Pasumbal' with '7 Active Loans' and navigation links: 'Home', 'All Payments', and 'People'. The form itself has a title 'New Loan Entry' and several input fields: 'Entry Name' (containing 'SNR Pizza'), 'Transaction Type' (a dropdown menu with 'Select Type'), 'Description' (containing 'pajutang pangkain lng ng mga junakis'), 'Date Borrowed' (containing '05/23/2026'), and 'Date Fully Paid' (a date picker showing 'mm/dd/yyyy').

Field	Required?	Description
Entry Name	Yes	A short descriptive label, eg. "Laptop Purchase Loan," "Friday KTV," "Office Lunch".
Transaction Type	Yes	Determines how the loan behaves. Choose one of the following options: • Straight Expense • Installment Expense • Group Expense
Description	No	Text to add more context or information.
Date Borrowed	No	Defaults to today. The calendar date the money was lent or spent.

Date Fully Paid	No	Leave blank on creation. This is set when the loan is closed. Cannot be earlier than Date Borrowed.
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Borrower and Lender

The screenshot shows a dark-themed form for creating a loan entry. The fields are as follows:

- Entry Name ***: Text input with value "Milktea".
- Transaction Type ***: Dropdown menu with value "Group Expense".
- Description**: Text input with placeholder "e.g. Loan for work accessories".
- Date Borrowed**: Date picker with value "05/23/2026".
- Date Fully Paid**: Date picker with placeholder "mm/dd/yyyy".
- Borrower Type ***: Radio buttons for "Person" (selected) and "Group".
- Borrower Group ***: Dropdown menu with value "Work Friends".
- Lender ***: Text input with value "David Jonathan Pasumbal".
- Amount Borrowed ***: Text input with value "500".
- Amount Remaining**: Text input with value "500".
- Payment Status**: Dropdown menu with value "UNPAID".
- Notes**: Text area at the bottom.

Field	Required?	Description
Borrower Type	Yes	Toggle between Person (individual borrower) or Group (shared expense among a group).
Borrower Person	Yes (if Person)	Name of the individual borrower
Borrower Group	Yes (if Group)	Select an existing group from the dropdown. Groups are managed under the People tab
Lender	Yes	The person who paid or lent the money. Defaults to "David Jonathan Pasumbal"

Financial Details

The screenshot shows a dark-themed form with three sections:

- Notes:** A text area with the placeholder text "Add loan details, conditions, notes..."
- Payment Notes:** A text area with the placeholder text "Add payment records, plans..."
- Receipt or Proof of Loan:** A section containing a "Choose File" button, the text "No file chosen", and a dashed box with the text "No receipt uploaded".

Field	Description
Notes	Text area for loan terms, conditions, or personal reminders
Payment Notes	Text area for payment plans or ad-hoc tracking notes
Receipt/Proof	Upload an image (photo of receipt, screenshot of a transfer).

Installment Plan Settings

When the **Transaction Type** is set to **Installment Expense**, an additional panel appears:

The screenshot shows a dropdown menu for "Transaction Type *". The menu is open, displaying the following options:

- Select Type
- Straight Expense
- Installment Expense
- Group Expense

The "Group Expense" option at the bottom is highlighted. Below the dropdown, the text "Date Fully Paid" is visible.

Field	Required?	Description
Start Date	Yes	The date of the first installment term. Defaults to the Date Borrowed if left blank
Payment Frequency	No	How often payments are due. Options: • Daily • Weekly • Bi-weekly • Monthly (default)
Number of Terms	No	How many installment cycles the loan is divided into.
Amount per Term	No	Auto-calculated. Equal to Amount Borrowed + Number of Terms . This field is read-only

(Note: The system uses flat-rate amortization (equal principal division). The amount per term is calculated by dividing the total principal evenly across all terms. There is no compounding interest calculation)

Payment Allocation Settings

When **Transaction Type** is set to **Group Expense**, a **Payment Allocation** panel appears:

Payment Allocation (Group Split)

Split Type

Split Equally
Split Equally
Split by Percent
Split by Amount

	Percentage (%)	Allocated Amount (₱)	Actions
Sabrina Ignacio	20.00	0.00	Remove
Ani Sibal	20.00	0.00	Remove
Razz Rios	20.00	0.00	Remove
Aisea Vidal	20.00	0.00	Remove
Shanessa Tugaoen	20.00	0.00	Remove

+ Add Group Member

Total Allocated: ₱0.00 / ₱0.00
✓ Split matches loan total.

Field	Description
Split Type	Split Equally: Each member's share is automatically calculated as $\text{Amount} \div \text{Number of Members}$. Both the percentage and amount fields will become read-only. Split by Percent: You manually enter each member's percentage. The system calculated the amount. Percentages must sum to exactly 100%. Split by Amount: You manually enter each member's amount. The system calculates the percentage. Amount must sum to the total loan amount.
Group Members Table	Lists each payee with their percentage, allocated amount, and status (UNPAID/PAID)
Auto-Population	When a Borrower Group is selected, the system will prompt to

	auto-populate the table with all the members of that group
Validation	A running total is displayed at the bottom of the table. The form cannot be submitted until the split balances correctly

Amortization Schedule

The amortization schedule applies only to installment expense loans

How the Schedule is Generated

When you save an installment loan, the program automatically generates individual terms based on your plan settings:

- **Term Count:** Equal to the **Number of Terms** you Specified
- **Due Dates:** Calculated from the **Start Date**, incremented by the **Payment Frequency** (e.g., monthly = +1 month per term)
- **Amount per Term:** Each term carries the same flat payment amount (**Principal + Terms**)

Viewing the Schedule

1. Open the loan by clicking on its card on the **All Payments Tab**.
2. Click the **Edit** icon to open the **Entry Detail Form**.
3. Scroll down to the **Installment Plan Settings** section.
4. Below the settings, the **Installment Terms Schedule** timeline is displayed, showing:
 - Term Number: e.g. Term 1, Term 2...
 - Due Date: The exact date the payment is expected.
 - Status Badge: The current status of each item.

Term Status

Field	Description
NOT STARTED	A future term that has not yet opened for payment.

UNPAID	The current active term, awaiting payment.
PAID	The term has been fully paid via recorded payments.
SKIPPED	The term was manually skipped. The amount appears with a strikethrough.
DELINQUENT	The due date has passed and no payment was recorded.

Approving/Activating a Loan

A loan's status is automatically managed through its payment activity.

1. On creation: A new loan is saved with the status **UNPAID** and **Amount Remaining** equal to the full amount borrowed. The loan is immediately active and visible on the **Home** dashboard.
2. First Payment Recorded: Once a payment is logged against the loan (via the **Payment Records** page), the status automatically changes to **PARTIALLY PAID** if the remaining balance is still greater than zero
3. Manual Override: In **Edit** mode, the **Payment Status** dropdown allows you to manually set the status to **UNPAID**, **PARTIALLY PAID**, or **PAID**. This is useful for correcting edge cases

Closing a Loan

A loan is considered closed when its outstanding balance reaches zero

Automatic Closure

- When recorded payments equal or exceed the total **Amount Borrowed**, the system sets **Amount Remaining** to 0.00 and transitions the **Payment Status** to **PAID**.
- The green progress bar on the **Payment Records** tab will fill to 100%
- For **Installment Expenses**, the loan is fully closed when all terms reach either **PAID** or **SKIPPED** status and the remaining balance is zero.

Setting the Closure Date

1. Open the loan in **Edit** mode.
2. Set the **Date Fully Paid** field to the date of the final payment
3. Click **Save**.

IMPORTANT: The **Date Fully Paid** field is not automatically set by the system. After a loan reaches **PAID** status, it is recommended to manually enter the closure date for your records

What Happens After Closure?

- The loan remains visible on the **All Payments** tab and can be filtered by the **PAID** status filter
- The loan's card on the **Home** dashboard will display a **PAID** badge.
- If a payment is later deleted from a closed loan, the status will revert to **PARTIALLY PAID** or **UNPAID**, effectively re-opening the loan.

Payment Processing

This section covers how to record routine transactions, handle irregular payment amounts, apply penalties, and process final payoffs.

Recording a Standard Payment

How to log a routine, on-time payment matching the scheduled amount. To record a standard payment toward an active loan, follow these steps:

1. Navigate to the **All Payments Records** page from the top navigation bar. Find the target transaction (e.g., KTV Music 21) and click the **View All** button on its row to expand its details.
2. Click the green **+ Add Payment** button to open the **New Payment** modal form.
3. Fill in the following mandatory fields:

Field	Required?	Description
Payment Date	Yes	The calendar date the payment was received. Defaults to the current system date.
Payment Amount (₱)	Yes	The numeric monetary amount received from the borrower. Must be greater than zero.
Payee Person	Yes	Dropdown selector to specify the user or group member providing the funds.
Payment Method	No	The transaction medium used. Options include: • Cash • Gcash • Bank Transfer
Notes	No	Free-text field for administrative remarks, reference codes, or transaction tracking details.

Proof of Payment	No	File upload field to attach images, screenshots, or PDF transaction receipts.
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4. Payment Amount (₱): Enter the exact cash amount received. Payee Person: Select the person making the payment from the dropdown list. Payment Method: Choose the transaction medium (e.g., Cash, Bank Transfer, GConnect). (Optional) Upload an attachment under Proof of Payment (e.g., a transaction receipt screenshot) and append any reference numbers under Notes. Click Submit. The loan's Remaining Amount will automatically deduct the payment, and the progress bar will dynamically recalculate toward 100%.

Handling Partial Payments

When a borrower makes a payment that is less than the expected amount, you should simply log the payment with the exact amount received.

For Straight or Group Expenses:

- The system will automatically deduct from the total **Amount Remaining**. The payment status of the loan will automatically update to "**PARTIALLY PAID**" to reflect that there is still an outstanding balance.

For Installment Plans:

- The system pools all payments made towards the loan. If the total payments made are not enough to cover the current installment term's required amount, the system will consume that partial amount but the specific term's status will remain as "**UNPAID**" (or "**DELINQUENT**" if it is past the due date). The term will only update to "**PAID**" once subsequent payments bring the total paid amount up to the required threshold for that term.

Troubleshooting and FAQs

anticipated roadblocks, common error messages, and troubleshooting guidelines for the Loan Tracking System:

Common Errors & Fixes

- **Error Message:** Failed to connect to backend service. Make sure the API server is running on `http://localhost:8080`
 - **What it means:** The frontend client cannot communicate with the Spring Boot backend REST API.
 - **How to fix:** Ensure the Spring Boot application is active and running. Go to your backend repository folder in terminal/cmd and run:

None

```
./mvnw spring-boot:run
```

Verify that your server successfully starts and binds to port 8080.

- **Error Message:** Borrower Person not found with ID... or Lender Person not found with ID...
 - **What it means:** The application is attempting to modify a loan associated with a Person or Group UUID that does not exist or has been deleted.
 - **How to fix:** Refresh the frontend page and select an existing contact/group from the directory.
- "Invalid Interest Rate" (or similar out-of-bounds inputs):
 - **What it means:** *Note: The current ERD specifications model interest-free loans (Straight, Group, or Installments without interest percentages). If custom extensions are implemented later and throw this error, it means the input is outside acceptable ranges (e.g. negative values).*
 - **How to fix:** Input only positive numeric numbers in monetary, amount, and term frequency inputs.

Getting Support

If you encounter system crashes, unresolvable validation bugs, or database constraint failures:

- **Internal Bug Tracking:** Submit an issue on the repository issue tracker.
- **Developer Contacts:** Reach out directly to the creators of the project
 - `sgignacio@up.edu.ph`
 - `rhrios@up.edu.ph`
 - `alsibal@up.edu.ph`
 - `sstugaoen@up.edu.ph`
 - `aevidal@up.edu.ph`